Physical Cume

September 18, 2009

1.) Summarize the quantization rules for rotational motion. (15 points)

2.) In molecular rotational spectroscopy, what are a) spherical rotors; b) symmetric rotors; and c) asymmetric rotors? (10 points)

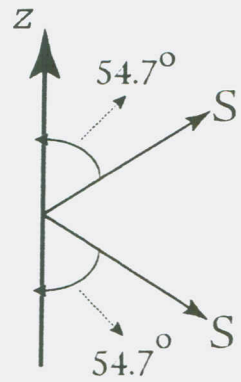
3.) Determine for each of the molecules in the list below, whether it is a spherical rotor, a symmetric rotor, or an asymmetric rotor (10 points):

a) CH_4 b) H_2O c) CH_3CN d) NH_3 e) SF_6 f) CH_3Cl g) CH_3OH

4.) State which of the following molecules have an observable pure rotational absorption spectrum (10 points):

a) N_2 b) CO_2 c) H_2O d) $CH_2{=}CH_2$ e) C_6H_6 f) CH_4 g) SF_6.

5.) State which of the following molecules have an observable rotational Raman spectrum (10 points):

a) N_2 b) CO_2 c) H_2O d) $CH_2{=}CH_2$ e) C_6H_6 f) CH_4 g) SF_6.

6.) For the specific case of the electron, show that the angle between the spin angular momentum vector, S, and a z-direction in the laboratory is 54.7^o as shown in the Figure below. (45 points)